PART 0 — ROADMAP & FOUNDATIONS

Campbell–Baker–Hausdorff

A Companion to Rossmann §1.3

The question that motivates everything:

$$\exp(X)\ \exp(Y) = \,?$$

> *"Never introduce a definition before the mathematical question that forces it."*
>
> — The guiding principle of these notes

§1

The Setup

Let G be a Lie group and $\mathfrak{g}$ its Lie algebra — the tangent space at the identity element $e \in G$. The exponential map sends each Lie algebra element to a group element.

X	Y
$X \in \mathfrak{g}$ *element of the Lie algebra*	$Y \in \mathfrak{g}$ *independent direction*
A = EXP(X) $A = \exp(X)$ *group element $A \in G$*	B = EXP(Y) $B = \exp(Y)$ *group element $B \in G$*

Their product $AB = \exp(X)\exp(Y)$ is again in G. The central question is: can this product itself be written as a single exponential?

$$\exp(X)\ \exp(Y) = \exp(Z)$$

...for some $Z \in \mathfrak{g}$? Everything in Rossmann §1.3 exists to answer this.

§2

The Optimistic Guess

If matrices behaved like ordinary numbers — if multiplication were commutative — we would naively expect:

$$\exp(X)\ \exp(Y) \stackrel{?}{=} \exp(X+Y)$$

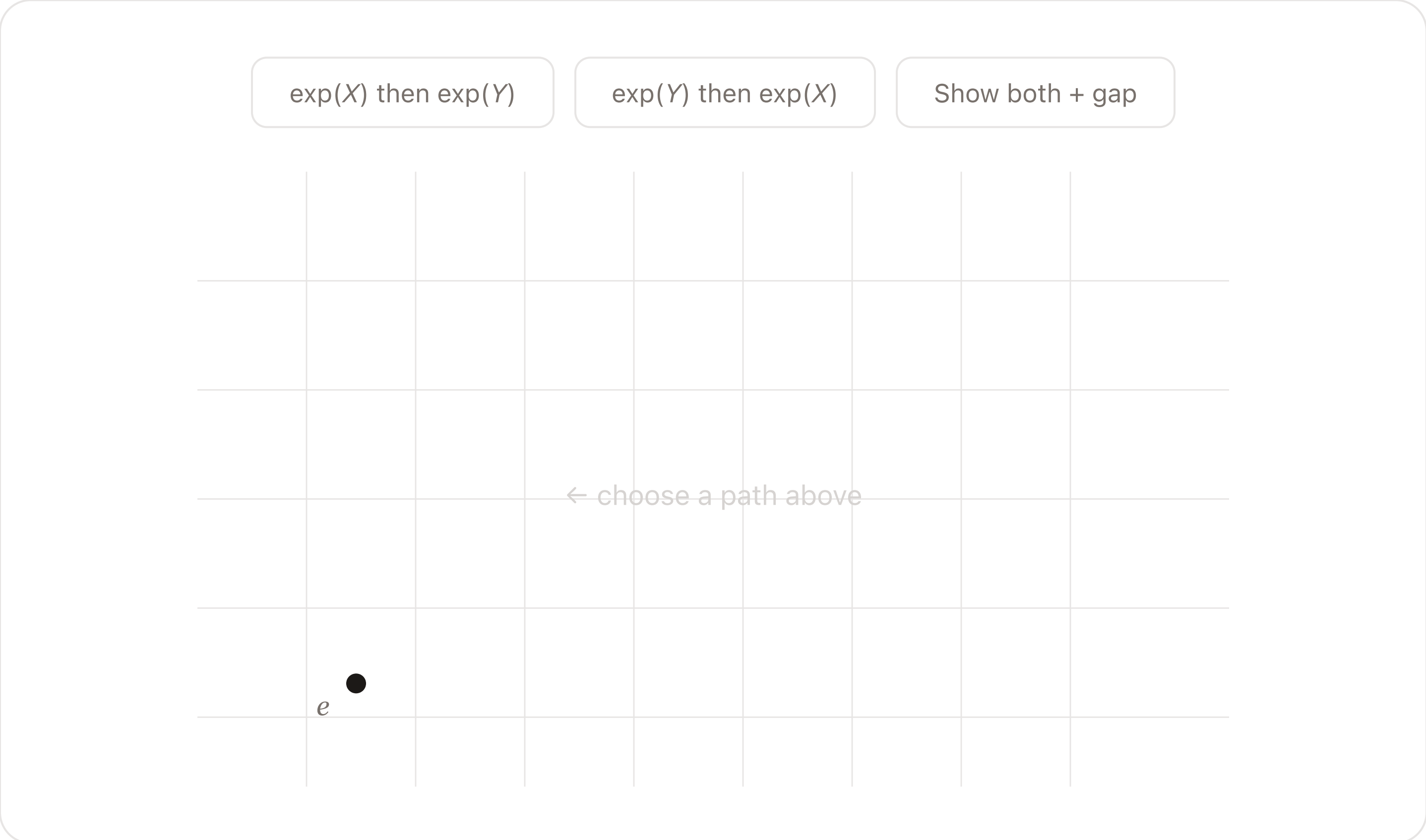
✓ HOLDS WHEN	× FAILS WHEN
$XY = YX$ *$XY = YX$ — matrices commute*	$XY \neq YX$ *the generic case*

The failure of commutativity is therefore the *entire source* of the difficulty — and of the interest.

§3

Non-commutativity is Geometric

The Lie group G is a curved manifold. On a curved space, the order in which you compose infinitesimal motions matters. Exponentiating X then Y lands you at a *different point* than Y then X.



The Lie bracket measures this failure to close. It is not an *arbitrary* algebraic structure — it is *forced* upon us by the geometry of composition.

§4

What the Logarithm Produces

Expanding $\log\big(\exp(X)\exp(Y)\big)$ via Taylor series produces ordinary matrix products:

$$\log\big(\exp(X)\exp(Y)\big) = X + Y + \cdots$$

First correction	$XY - YX$
Second order	$X^2Y,\ XY^2,\ YX^2,\ \ldots$
And more...	$XYX,\ X^2Y^2,\ XYXY,\ \ldots$

"Nothing yet resembles Lie theory."

→ But collect the terms. Something remarkable happens.

§5 ✦

The Lie Bracket Appears

When we collect the first correction terms, the ordinary matrix product $XY - YX$ is exactly the commutator:

✦ THE FIRST MIRACLE

$$XY - YX \quad = \quad [X, Y]$$

See the first correction →

The complete BCH series, due to Dynkin, expresses Z entirely in terms of nested Lie brackets:

$$Z = X + Y + \tfrac{1}{2}[X,Y] + \tfrac{1}{12}[X,[X,Y]] - \tfrac{1}{12}[Y,[X,Y]] + \cdots$$

§6

The Philosophical Meaning

The bracket $[X, Y]$ measures the failure of infinitesimal motions to commute.

Move first by X, then by Y. Reverse the order. The gap between where you end up is measured by $[X, Y]$.

> *"[X,Y] is not an arbitrary algebraic definition — it is forced upon us by the geometry of composition."*

Any attempt to understand how group elements compose in terms of infinitesimal data will inevitably produce $[X, Y]$. The Lie bracket is *necessary*, not chosen.

§7

Why Dynkin's Formula is Deep

Rossmann proves that *every* correction term in the BCH series can be written solely using nested Lie brackets. No ordinary matrix products remain:

Nested brackets in the series:

$[X, Y]$

$[X,[X,Y]]\ ,\ [Y,[X,Y]]$

$[X,[X,[X,Y]]]\ ,\ \ldots$

No bare product XY survives.

The local multiplication law of the Lie group is *completely encoded* by the Lie algebra and its bracket. The group and the algebra are, locally, the same structure.

§8

Roadmap for the Full Companion

The planned full notes (estimated 80–120 pages) will cover:

01 Motivation from geometry

02 Derivation of BCH from first principles

03 Complete expansion of logarithm and exponential

04 Why commutators inevitably appear

05 Dynkin's formula explained symbol-by-symbol

06 Differential-equation proof line by line

07 The operators ad, γ and δ with intuition

08 Historical development

09 Fully worked Rossmann problems

THE ANSWER

$$\exp(X)\exp(Y) = \exp\big(X + Y + \tfrac{1}{2}[X,Y] + \cdots\big)$$

End of Part 0.

Part II: Derivation from first principles →